

DIAGRAMS

AUTHENTICATION SYSTEM

REGISTRATION FLOW

1. Client sends: `POST /api/auth/register { name, email, password, studentId, department }`
2. Server validates: required fields, email format, password ≥ 6 chars, unique email
3. Mongoose pre-save hook hashes password with bcrypt (cost=12)
4. User document saved to MongoDB $\rightarrow$ JWT signed $\rightarrow$ { token, user } returned (201)

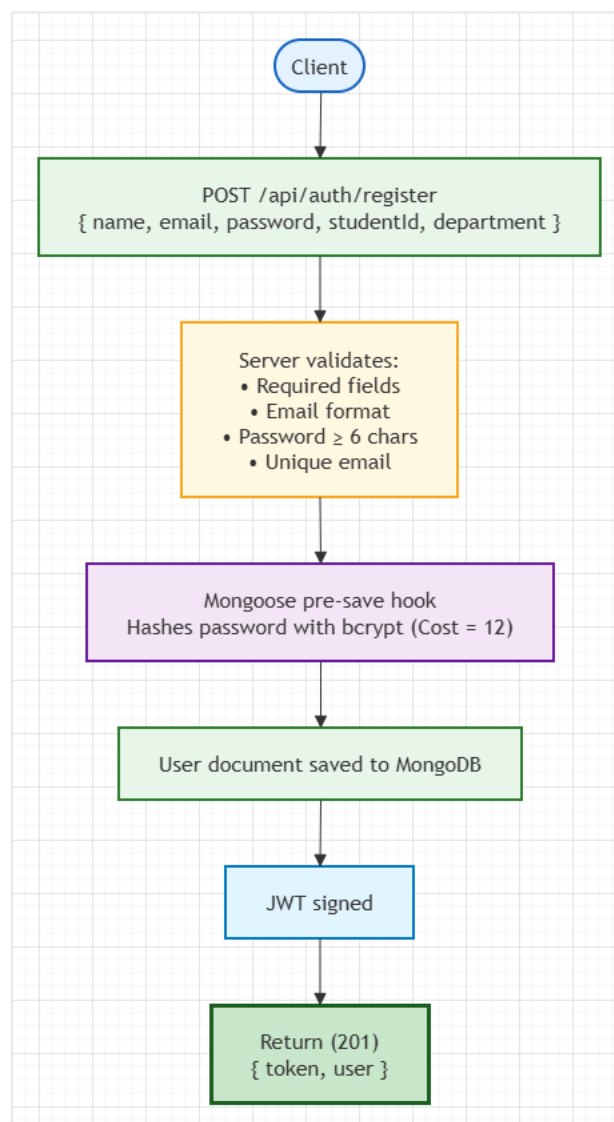


Figure : Registration Flow

LOGIN FLOW

1. Client sends: `POST /api/auth/login { email, password }`
2. Server finds user by email (with password field selected)
3. `bcrypt.compare()` validates password against stored hash
4. `lastLogin` timestamp updated → JWT issued → `{ token, user }` returned (200)

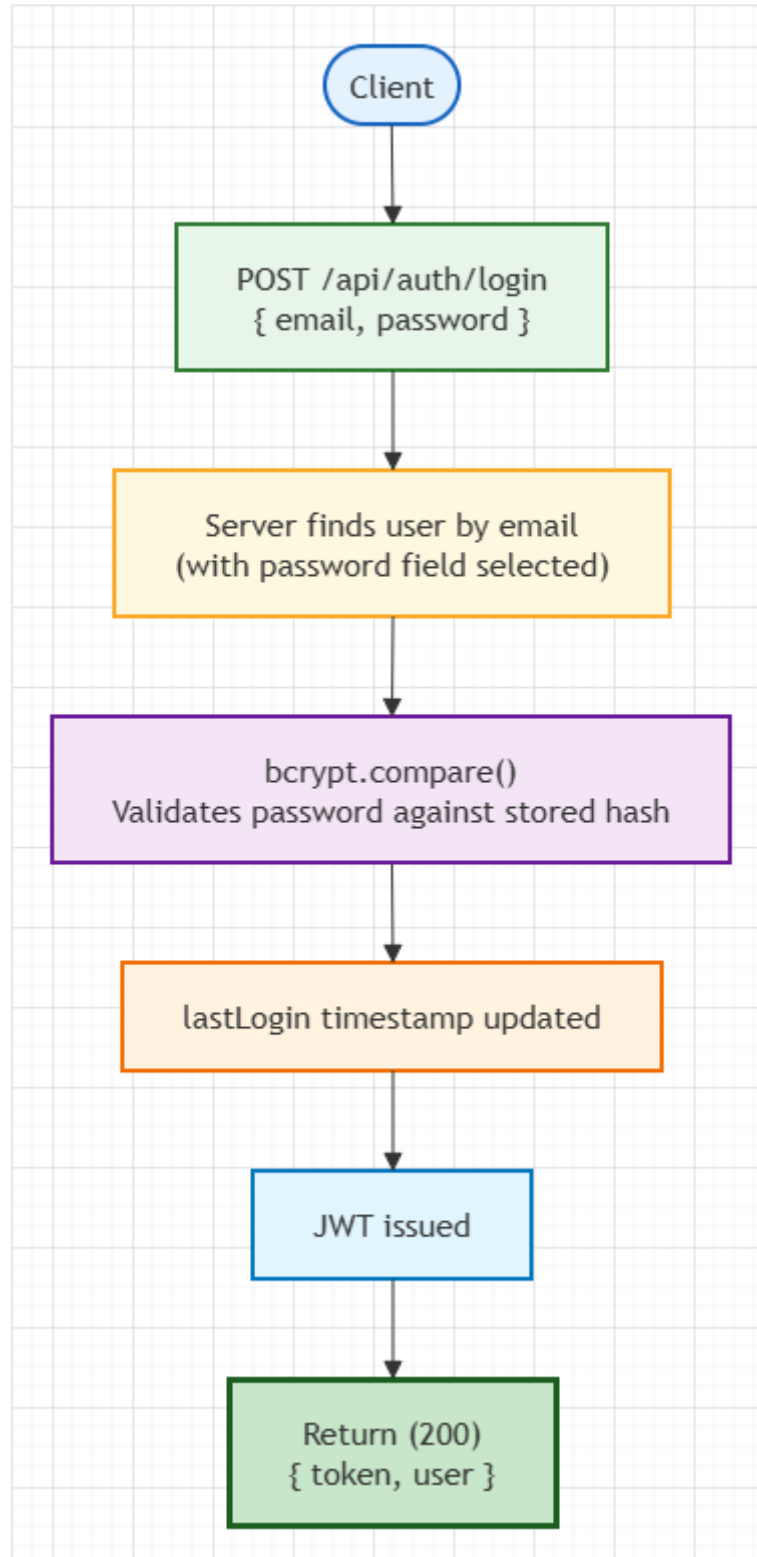


Figure : Login Flow

PROTECTED ROUTE ACCESS

1. Client sends: Authorization: Bearer <token> in request header
2. auth middleware extracts token, calls jwt.verify() with JWT_SECRET
3. Decoded payload { id, email, role } attached to req.user
4. For admin routes: adminOnly middleware checks req.user.role === 'admin'

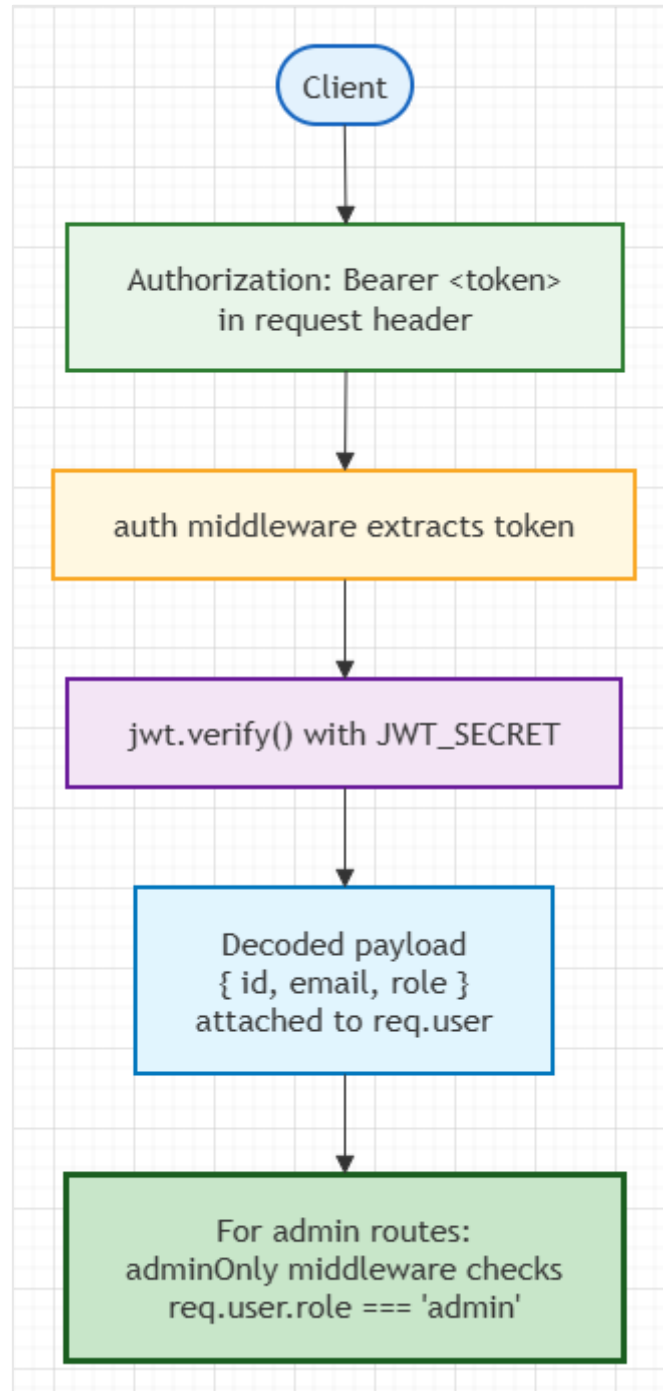


Figure : Protected Route Access Flow